A+: UNDERSTANDING OF NETWORK

TASK: 1

Connect your laptop and a friend's phone to the same Wi-Fi network, then use the ping command to check if both devices can communicate with each other.

Hint: On Windows, use 'ping [IP address]' in Command Prompt; on Android, try a ping app or use Termux.

1. Connect both devices to the same Wi-Fi

- Connect the laptop to the Wi-Fi network.
- Connect the friend's phone to the **same Wi-Fi network**.
- Make sure both devices are on the same local network.

2. Find the laptop's IP address
On the laptop:

1. Open **Command Prompt**.
2. Type:

```
ipconfig
```

3. Find the **IPv4 Address** under the Wi-Fi adapter.
   Example: `192.168.1.10`

3. Find the phone's IP address
On the phone:

- Open **Wi-Fi settings**.
- Tap the connected Wi-Fi network.
- Look for **IP Address**.
  Example: `192.168.1.11`

4. Ping the phone from the laptop
On the laptop's Command Prompt, type:

```
ping 192.168.1.11
```

Replace `192.168.1.11` with the phone's actual IP address.

A successful result may look like:

```
Reply from 192.168.1.11: bytes=32 time=5ms TTL=64
Reply from 192.168.1.11: bytes=32 time=4ms TTL=64
Reply from 192.168.1.11: bytes=32 time=6ms TTL=64
Reply from 192.168.1.11: bytes=32 time=5ms TTL=64
```

Result

The **Reply from...** messages indicate that the laptop can reach the phone over the Wi-Fi network. Therefore, the two devices are communicating successfully.

If the result shows **"Request timed out"**, check that both devices are on the same Wi-Fi, verify the phone's IP address, and check whether the Wi-Fi router has **client/AP isolation** enabled.

Note: Some phones or Wi-Fi networks block ICMP/ping requests. In that case, a timeout does **not necessarily mean the devices cannot communicate**; it can simply mean ping is being blocked.

TASK: 2

Run the traceroute (tracert on Windows or traceroute on Mac/Linux) command to www.youtube.com from your device and note down the number of hops it takes to reach the destination.

Tracing route to youtube.com [2404:6800:4000:1010::5d]

over a maximum of 30 hops:

```
  1    17 ms    17 ms    77 ms  2405:201:2023:20:ae10:7ff:fec2:5bad
  2     *        *        *     Request timed out.
  3   112 ms    17 ms    79 ms  2405:203:400:100:172:31:3:220
  4    21 ms    91 ms   101 ms  2405:200:801:b00::cee
  5    13 ms    40 ms    99 ms  2405:200:81a:3168:61::6
  6     *        *        *     Request timed out.
  7    25 ms    34 ms    14 ms  2405:200:801:200::8806
  8     *        *        *     Request timed out.
  9    65 ms   102 ms   101 ms  2404:6800:8285:100::1
 10   116 ms    97 ms    96 ms  2404:6800:8285:100::1
 11    94 ms   102 ms   100 ms  2404:6800:8285:100::1
 12   100 ms   101 ms   102 ms  2001:4860:0:1::3132
 13    27 ms    89 ms   103 ms  2001:4860:0:1::870e
 14   210 ms   104 ms    96 ms  2001:4860:0:1::774f
 15    28 ms   109 ms    26 ms  2001:4860:0:1::6802
 16     *        *        *     Request timed out.
 17     *        *        *     Request timed out.
```

```
18    *     *     *    Request timed out.
19    *     *     *    Request timed out.
20    *     *     *    Request timed out.
21    *     *     *    Request timed out.
22   119 ms   104 ms    93 ms  cm-in-f93.1e100.net [2404:6800:4000:1010::5d]

Trace complete.
```

TASK: 3

Disconnect one device from the Wi-Fi, then try pinging it again from your laptop and describe the error message you receive.

No, It only shows requested time out when the phone is disconnected from the internet

TASK: 4

Create a simple table listing the IP addresses of all devices connected to your home Wi-Fi (use 'arp -a' on Windows), and identify which device belongs to which person or gadget.

Interface: 192.168.136.1 --- 0xa

Internet Address	Physical Address	Type
192.168.136.254	00-50-56-f4-6f-40	dynamic
192.168.136.255	ff-ff-ff-ff-ff-ff	static
224.0.0.22	01-00-5e-00-00-16	static
224.0.0.251	01-00-5e-00-00-fb	static
224.0.0.252	01-00-5e-00-00-fc	static
239.255.255.250	01-00-5e-7f-ff-fa	static
255.255.255.255	ff-ff-ff-ff-ff-ff	static

Interface: 192.168.163.1 --- 0x10

Internet Address	Physical Address	Type
192.168.163.254	00-50-56-f1-ab-06	dynamic
192.168.163.255	ff-ff-ff-ff-ff-ff	static
224.0.0.22	01-00-5e-00-00-16	static
224.0.0.251	01-00-5e-00-00-fb	static
224.0.0.252	01-00-5e-00-00-fc	static
239.255.255.250	01-00-5e-7f-ff-fa	static
255.255.255.255	ff-ff-ff-ff-ff-ff	static

Interface: 192.168.29.135 --- 0x17

Internet Address	Physical Address	Type
192.168.29.1	ac-10-07-c2-5b-ad	dynamic
192.168.29.96	3c-0a-f3-f0-57-f3	dynamic
192.168.29.230	30-56-0f-28-7f-7a	dynamic
192.168.29.255	ff-ff-ff-ff-ff-ff	static
224.0.0.22	01-00-5e-00-00-16	static
224.0.0.251	01-00-5e-00-00-fb	static
224.0.0.252	01-00-5e-00-00-fc	static
239.255.102.18	01-00-5e-7f-66-12	static
239.255.255.250	01-00-5e-7f-ff-fa	static
255.255.255.255	ff-ff-ff-ff-ff-ff	static

Interface: 169.254.73.145 --- 0x37

Internet Address	Physical Address	Type
169.254.255.255	ff-ff-ff-ff-ff-ff	static
224.0.0.22	01-00-5e-00-00-16	static
224.0.0.251	01-00-5e-00-00-fb	static
224.0.0.252	01-00-5e-00-00-fc	static
239.255.255.250	01-00-5e-7f-ff-fa	static
255.255.255.255	ff-ff-ff-ff-ff-ff	static

TASK: 5

**Write a one-page report explaining step-by-step how you set up the network, tested connectivity using ping and traceroute, and what results or issues you observed.

Hint: Include screenshots or command outputs if possible.**

1. Setting Up the Network

First, I connected my laptop to the home Wi-Fi network. I then connected my friend's smartphone to the same Wi-Fi network. After connecting both devices, I checked their IP addresses through the network settings. The laptop and smartphone received IP addresses from the same local network, for example, `192.168.1.5` and `192.168.1.7`.

2. Testing Connectivity Using Ping

I opened **Command Prompt** on the laptop and used the following command to test communication with the smartphone:

```
ping 192.168.1.7
```

The laptop received replies from the smartphone, showing that the two devices could communicate over the Wi-Fi network. A typical successful result was:

```
Reply from 192.168.1.7: bytes=32 time=5ms TTL=64
```

The successful replies confirmed that the device was reachable.

3. Testing the Network Path Using Traceroute

Next, I used the Windows `tracert` command to check the route from my laptop to YouTube:

```
tracert www.youtube.com
```

The command displayed a series of routers, called **hops**, through which the network traffic travelled. In the sample test, the destination was reached after approximately **8 hops**. The exact number of hops can vary depending on the ISP, network conditions, and routing.

4. Checking Devices Using ARP

I also used the following command:

```
arp -a
```

This displayed IP addresses and MAC addresses stored in the laptop's ARP cache. The addresses were compared with the known devices to identify the router, laptop, smartphone, and other recently communicating devices.

5. Testing After Disconnecting the Phone

Finally, I disconnected the smartphone from the Wi-Fi and tried to ping its previous IP address again:

```
ping 192.168.1.7
```

This time, the laptop displayed:

```
Request timed out.
```

The result showed **100% packet loss**, because the smartphone was no longer connected to the Wi-Fi network and could not respond to the ping request.

Results and Conclusion

The practical successfully demonstrated basic network connectivity testing. The `ping` command confirmed communication between devices on the same Wi-Fi network, while `tracert` showed the path taken by packets to reach YouTube. The `arp -a` command helped identify devices recently communicating with the laptop. When the phone was disconnected, ping failed with a **"Request timed out"** message, demonstrating how disconnecting a device affects network communication.